



Microbial Nitrogen Removal in South San Francisco Bay: Does It Play a Role in Eutrophication Resistance?

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Abstract

The ecosystem response to anthropogenic nitrogen (N) loading in estuarine systems is determined by hydrodynamics, biogeochemical transformation rates, and other system-specific characteristics. Historically, San Francisco (SF) Bay has been an outlier from other estuaries with an unusual resistance to eutrophication, despite having extremely high rates of nitrogen loading. Recent increases in phytoplankton biomass and an unprecedented harmful algal bloom, however, have increased the urgency to understand rates and drivers of nitrogen removal in the system. To assess benthic N cycling rates, we conducted seasonal measurements across nine sites in South and Lower South SF Bay, the two sub-embayments with the highest rates of area-normalized N loading, to determine the rates and potential drivers of denitrification and dissimilatory nitrate reduction to ammonium (DNRA). Denitrification rates averaged $60.6 \pm 8.1 \mu\text{mol m}^{-2} \text{h}^{-1}$ and were primarily coupled to nitrification. Denitrification rates were positively correlated with DNRA rates and % clay. DNRA rates ranged from 0 to $20 \mu\text{mol m}^{-2} \text{h}^{-1}$ and on an annual basis averaged ~10% of total benthic nitrate reduction, with a negative correlation to % clay content in the surface sediment. The measured denitrification rates account for the removal of, on average, 14% of N loaded annually to South SF Bay, leaving a sizeable portion for alternate fates (e.g., recycling, export, or burial) and potential for substantial temporal and spatial variability (1–79%). This identifies the relative importance of sediment denitrification in ecosystems characterized by high nutrients and low productivity.

Keywords Nitrogen loading · Denitrification · Estuary · Eutrophication · Productivity · Dissimilatory nitrate reduction to ammonium

Introduction

High anthropogenic nitrogen (N) loading rates impact many densely populated coastal ecosystems worldwide (Howarth et al. 2002; Malone & Newton 2020). Excess N inputs in estuarine systems often lead to high rates of

primary production and eutrophication-related effects such as hypoxia, toxic algal blooms, and fish kills (Nixon et al., 1995). However, some estuaries appear to be less vulnerable to the negative impacts of eutrophication. For example, San Francisco (SF) Bay historically has shown little evidence of eutrophication, despite having one of the largest annual N loads relative to all estuaries globally, primarily from wastewater treatment plants surrounding the Bay (Cloern et al. 2020). However, with population growth, increased N loading, and local climate change, the Bay appears increasingly susceptible to eutrophication (Cloern et al. 2020) and its negative effects, as exhibited by an unprecedented harmful algal bloom event in August 2022 (Kudela et al. 2023). Although it is recognized that high suspended sediment concentrations in South SF Bay limit light penetration, restrict algal growth, and mute eutrophication effects (Painting et al. 2007), the importance of microbial N removal and other N cycling processes has

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been poorly characterized. It is critical to understand the Bay's N cycling dynamics and removal of N inputs to predict the impact of future changes on water quality and inform managers how to mitigate impacts.

Because the health of San Francisco Bay is important to the quality of life of approximately 7 million people in the highly urbanized region surrounding the Bay, understanding the biogeochemical dynamics that maintain its health under extreme N loading conditions is critical. South SF Bay is a shallow lagoon-like estuary without significant natural freshwater inputs. N loading rates to the South Bay averaged 19.2 metric tons N per day between 2012 and 2020, primarily from wastewater effluent released by several publicly owned treatment works (POTWs). Despite high nutrient availability, productivity in SF Bay has been historically low relative to other estuaries (Jassby et al. 2002; Kimmerer et al. 2012), with the Bay typically experiencing spring phytoplankton blooms with highly variable annual biomass (Cloern et al. 2020; Beck et al. 2022). High concentrations of resuspended sediments in South SF Bay (Damashek et al. 2016) and strong tidal mixing (Cloern et al., 1991) limit algal production and deposition except under highly stratified conditions (Cloern 1987). Natural N removal processes performed by microbes can play a critical role in protecting estuarine systems from eutrophication, but this has not yet been assessed in South SF Bay.

Denitrification and dissimilatory nitrate reduction to ammonium (DNRA) are two major microbial metabolic processes that compete for nitrate (NO_3^-) electron receptors in benthic estuarine sediments. Whereas denitrification is a primary N removal process in many estuaries, anammox (anaerobic ammonium oxidation) typically plays a lesser role (Damashek & Francis 2018, Gao et al. 2022, Trimmer and Nicholls 2009, Wang et al. 2012). Denitrification is often coupled to nitrification, an aerobic microbial process that oxidizes ammonia (NH_3) to NO_3^- . Spatial and temporal variation of estuarine denitrification rates may be driven by or correlated with characteristics such as temperature, NO_3^- availability, sediment oxygen demand (Cornwell et al. 1999), sediment organic matter (Caffrey et al. 1993), and grain size (Gongol et al., 2016). DNRA, an often-ignored N cycling pathway, competes for NO_3^- with denitrification and may impact denitrification rates in environments where temperature, sulfide, and labile OC are high (Burgin & Hamilton 2007; Giblin et al. 2013). Unfortunately, because the majority of studies of DNRA have been done with slurries rather than intact cores, little is known about DNRA function in estuarine systems with natural sediment redox gradients. Slurry rate measurements may overestimate DNRA rates happening in the natural environment because they mix deeper,

more reduced sediment with oxidized surface sediment. Here we add to the limited studies (Christensen et al. 2000; Dong et al. 2011; Roberts et al. 2014; Marchant et al. 2016; Magri et al. 2022) assessing estuarine DNRA rates measured in intact sediment cores that reflect more realistic in situ sediment redox gradients.

Although paradigms of estuarine biogeochemistry have been well established over decades of studies, these paradigms may not apply to estuaries everywhere but tend to be biased toward a more "classical" estuary model. Early studies on estuaries such as the Chesapeake or Narragansett Bays began to shape the paradigms of estuarine science (e.g., Nixon, Paerl, Howarth, Kemp). A classical estuary model is represented by an upstream to downstream gradient of salinity from low to high and nutrients from high to low, often leading to gradients of other ecosystem characteristics such as metabolic processes and nutrient transformation rates (Hopkinson et al. 1999; Boynton & Kemp 1985). Because N availability limits productivity in most marine systems (Paerl et al., 2018), increasing N typically stimulates productivity. The organic matter produced by increased productivity often stimulates aerobic respiration, depleting oxygen in deeper water, and stimulating anaerobic processes such as denitrification. However, atypical estuaries such as South SF Bay lack many of the attributes of this estuarine model, and therefore the classical estuarine paradigms may not apply. South SF Bay does not have classical estuarine gradients in salinity, nutrients, and suspended sediments, and rather than the typical estuarine N and sometimes P limitation of productivity (Paerl et al., 2014), production is low and limited by light (Cloern et al. 2020). Nutrient gradients in the Bay are driven by several POTW point sources rather than upstream freshwater nutrient inputs. Blooms in estuaries are typically initiated by warmer temperatures, stratification, and high N inputs (Carstensen et al. 2007). Because the Bay is tidally well-mixed, stratification tends to be short-lived and dissolved oxygen is rarely depleted. Thus, with these unique attributes, South SF Bay would not be expected to function biogeochemically as a typical estuary.

Measurements of sediment nitrogen cycling rates in San Francisco Bay are few and reflect different conditions and nutrient loading rates in the Bay when they were first measured in the 1980s and 1990s. Observations of nutrient fluxes from 1991 to 1993 were spatially consistent across the South SF Bay (Caffrey et al. 1996). Denitrification within intact sediment cores from the north SF Bay Delta region (Cornwell et al. 2014), potential denitrification rates in sediment slurries (Mosier & Francis 2010), and intertidal sediment denitrification measurements in a slough in South San Francisco Bay (Oremland et al. 1984), represent the most current data for denitrification in SF Bay. Understanding

the drivers of N recycling and removal processes in low-productivity systems may improve annual N removal estimates at a global scale and inform model predictions of rates of N removal processes associated with climate change, sea level rise, changes to anthropogenic N loading, and other potential shifts in local physical/chemical characteristics.

The goal of this study was to determine rates of N cycling and removal dynamics in South San Francisco Bay as a model for estuaries with high sediment and nitrogen loads combined with low productivity. Although the San Francisco Bay is characterized by a sprawling and complex network of embayments, we focused our study specifically on the two major SF Bay sub-embayments that comprise South SF Bay, the South Bay and Lower South Bay, because they have the highest DIN loading rates (SFEI 2014) and highest DIN concentrations (SFEI 2014, Crauder et al. 2016) within the entire Bay. Here we introduce the first denitrification rates measured in benthic sediment cores in the heavily N-impacted South and Lower South SF Bay, and we ask the following questions:

- 1) What are the spatial and seasonal patterns of NO_3^- reduction rates (denitrification and DNRA) in South and Lower South San Francisco Bay?
- 2) What are the proximal and system controls of the spatial and temporal variation of NO_3^- reduction rates across South and Lower South San Francisco Bay?
- 3) Is N removal via denitrification an important fate for N loaded to the South and Lower South San Francisco Bay?

Based on the previously established paradigms of classical estuarine biogeochemistry, we hypothesized that South SF Bay would have an upstream to downstream gradient of NO_3^- which would result in higher NO_3^- availability and denitrification rates in upstream locations. We may observe higher rates of denitrification in the deep channel where stratification, which occurs intermittently during combinations of neap tides, high temperatures, low winds, and/or greater freshwater inputs, may limit oxygen availability and stimulate anaerobic metabolic processes. Although NO_3^- concentrations are typically highest in the water column during winter when phytoplankton nutrient uptake and growth is slow, denitrification is expected to be highest during the summer when temperature and respiration rates are high, labile organic matter is readily available from settled organic matter from spring production, and water column oxygen levels are lower. We hypothesized that denitrification rates would be greater than rates in typical estuaries due to greater-than-average NO_3^- availability. However, due to the atypical attributes of South SF Bay previously highlighted, we hypothesized that South SF Bay may deviate significantly from the paradigms of classical estuaries.

Methods

Site Description

South San Francisco Bay (Fig. 1) is a large lagoon-like embayment with typical salinities between 28 and 35, reflecting low freshwater inputs. While a 7- to 13-m deep channel runs along the length of the Bay, the vast majority of the Bay is < 5 m deep and the southern region is divided into two major sub-embayments: South Bay and Lower South Bay. Tidal range averages about 2.5 m, and water temperature typically ranges from 9 to 21 °C throughout the year. Water column nitrate concentrations range widely up to 120 μM and fluctuate with the tidal cycle, typically greater at high tide and lower at low tide (Fig. 2). Lower South Bay is surrounded by a significant area of salt ponds that have unknown water exchange rates with Lower South Bay.

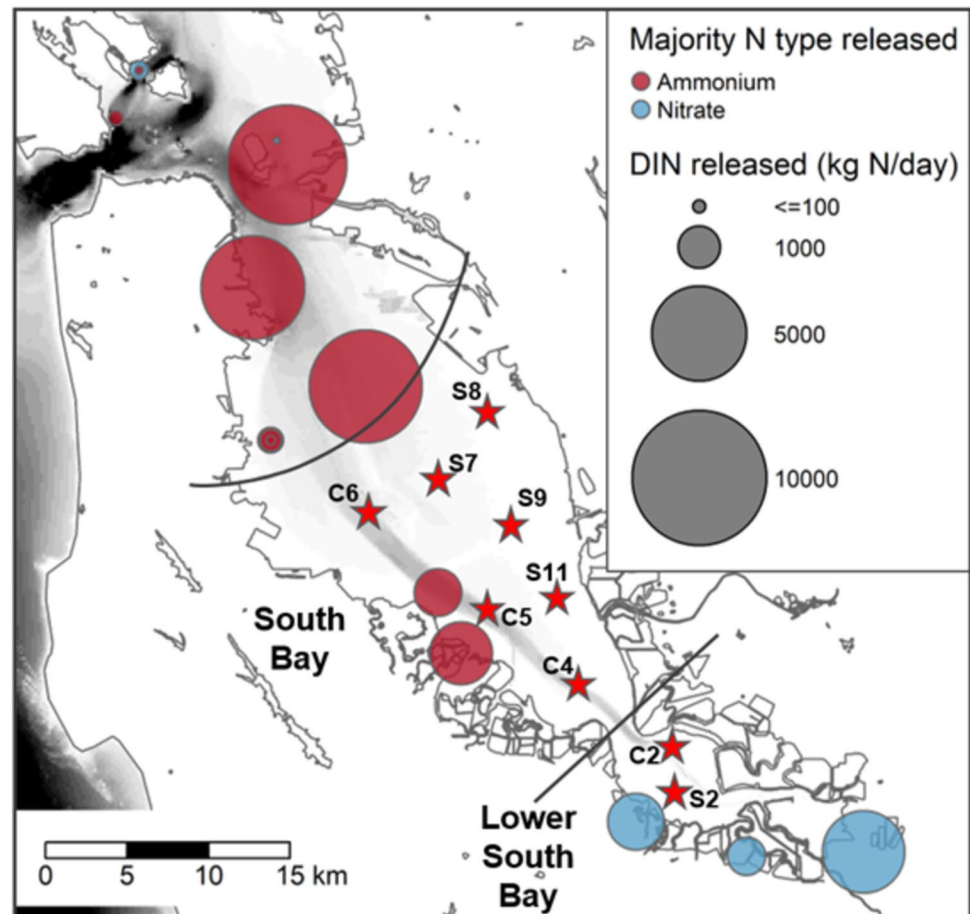
Sampling Design

To survey spatial variation in denitrification rates across seasons, nine sites were sampled in South San Francisco Bay. Four “channel” sites (C2, C4, C5, and C6, ordered from south to north) were established in the deep channel and corresponded to previously established USGS monitoring sites (35, 32, 30, and 27, respectively, Schraga & Cloern 2017). Five “shoal” sites (S2, S11, S9, S7, S8, ordered from south to north) were located in the broader shoal areas of the South Bay and Lower South Bay (Fig. 1). These nine sites were sampled seasonally using the S/V *Retriever* and the R/V *Jewell* (USGS) in February, May, August, and November 2021.

Incubations

At each site, duplicate sediment cores were collected for sediment–water exchange incubations, and additional cores were collected for isotope pairing technique (IPT) incubations for measurements of denitrification and DNRA. A GMX-25 GOMEX box corer (Ocean Instruments) or a light-weight 15 × 15 cm corer (Cornwell et al. 2016) were used to sample deeper channel sites and a pole corer was used to collect cores from shoal sites. The 28-cm-high 6.4- or 10-cm diameter core tubes were ~ 1/2 filled with sediment. DIN was measured in 0.45 μm pore-size-filtered bottom water at each site with a WestCo discrete nutrient analyzer. Upon return to the laboratory, cores were bubbled overnight in dark, aerated site water baths at the in situ water temperature measured at the time of collection. Two empty cores were filled with site water and served as “blank” cores alongside the sediment cores. The February, May, August, and November 2021

Fig. 1 Map of South San Francisco Bay showing seasonal sampling sites (red stars) and nitrate and ammonium point source release across the Bay (circles). Black lines represent boundaries of the two major embayments of South Bay and Lower South Bay. Sites that begin with “C” are deep channel sites, and sites that begin with “S” are shoal sites. DIN loading data are from water year 2021. Sites C2, C4, C5, and C6 correspond to USGS stations 35, 32, 30, and 27, respectively

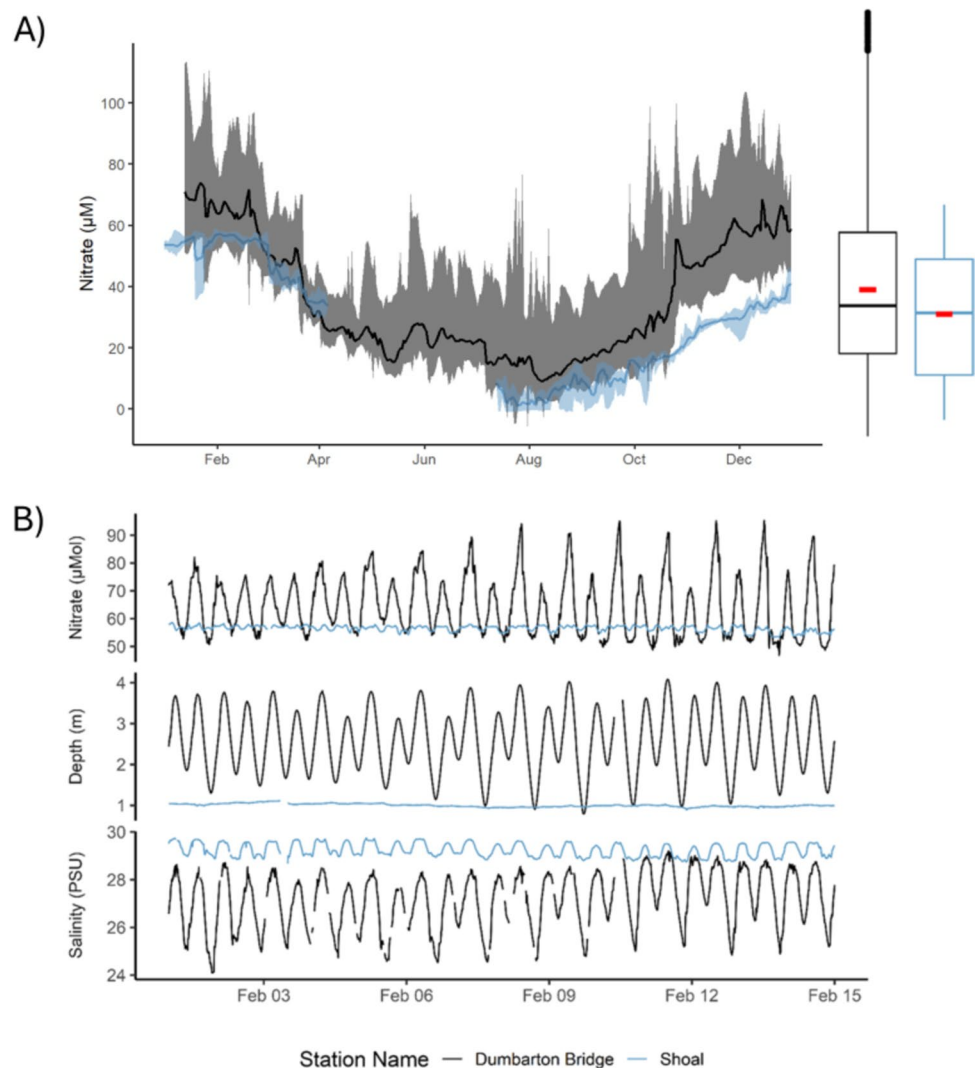


incubations were set to 13, 20, 22, and 17 °C, respectively, to match average bottom water temperatures measured at the sites on the day of sampling. The morning following core collection, flux cores were capped with magnetic stirring tops and incubated in the dark as described in Cornwell et al. (2016). Water overlying each core was sampled every 1–2 h for dissolved oxygen (DO), until DO was reduced to 70–80% of saturation. Overlying water was replaced with site water as sampling took place. Because less than 5% of the core water volume was removed per sampling, replacement with slightly more oxic overlying water had a negligible effect on the flux calculation. Dissolved oxygen was measured immediately using a PyroScience FireSting O₂ sensor. After flux incubations, the cores were subsampled for analysis of bulk density, organic matter content, grain size, and benthic chlorophyll.

A separate isotope pairing technique (IPT; Nielsen 1992) core incubation using $^{15}\text{NO}_3^-$ addition to trace production of $^{29}\text{N}_2$ and $^{30}\text{N}_2$, (to determine denitrification rates), and $^{15}\text{NH}_4^+$ (to determine DNRA rates) was run simultaneously in a different bath of aerated site water at in situ temperature in the dark as described in Murphy et al. (2016). One core from at least six different sites for each sampling event was

designated “ T_0 ” cores for measurement of initial $^{29}\text{N}_2$ and $^{30}\text{N}_2$ abundance, and cores from every site were designated “ T_f ” cores for measurement of production after the incubation. The water bath was spiked with $^{15}\text{N-KNO}_3$ tracer to a target concentration of 100 μM $^{15}\text{NO}_3^-$ and allowed to mix and diffuse into sediments for 2 h. T_f cores were then capped and T_0 cores were quickly slurried while minimizing bubbles and atmospheric exposure. Samples were immediately siphoned from slurries into 12-mL Exetainers™ and spiked with 70 μL saturated ZnCl_2 solution, then stored underwater at 4 °C until measurement of $^{29}\text{N}_2$ and $^{30}\text{N}_2$ production on a membrane inlet mass spectrometer (MIMS, Bay Instruments LLC). One hundred twenty milliliters of slurry was then sampled into a Whirlpak bag containing KCl salt to extract NH_4^+ for measurement of DNRA. DNRA samples were shaken for 1 h, filtered at 0.45 μm pore size, and poured into two Exetainers per sample. One Exetainer was spiked with hypobromite solution to oxidize NH_4^+ to N_2 and both Exetainers were analyzed on a MIMS (Yin et al. 2014). T_f cores were slurried and sampled after 20–30% of dissolved O₂ was depleted in the separate flux cores. Denitrification calculations followed the equations in Nielsen (1992) and DNRA calculations follow the equations in Yin

Fig. 2 Water column nitrate concentrations at Dumbarton Bridge (black) in the channel (located at the boundary between Lower South Bay and South Bay) and shoal site S9 (blue). **A** shows nitrate concentrations at daily average (dark line) and daily max/min range (lighter shading) over the year of sampling (2021, shoal data missing during spring/summer period due to lost sensor), with a box plot insert showing the differences in average nitrate concentrations (the red line is the mean). While the daily average concentrations are not different, the range is much greater for Dumbarton Bridge, frequently exposing the sediment to much higher nitrate concentrations. **B** focuses on a shorter time scale, showing patterns of nitrate concentrations and salinity with depth at Dumbarton Bridge (black) and S9 shoal site (blue) over several tidal cycles. Shoal site depth reading was constant because the sensor was attached to a floating buoy



et al. 2014. D15 represents the denitrification rate measured with $^{15}\text{NO}_3^-$ tracer, and thus is a “potential” rate, and D14 represents the “natural” denitrification rate measured if the ^{15}N labeled tracer was not present. Measurement of $^{29}\text{N}_2$ and $^{30}\text{N}_2$ production with increasing $^{15}\text{NO}_3^-$ spike concentrations confirmed that the key IPT requirements (e.g., $^{15}\text{NO}_3^-$ addition does not alter rates of denitrification, uniform mixing of $^{15}\text{NO}_3^-$, and no interference of high anammox rates with IPT calculations) (Eyre et al. 2002; Risgaard-Petersen et al. 2003) were met (see Supplemental Materials, Figure S1).

Sediment Characteristics

After the incubation, the top 2 cm of sediment from each flux core was subsampled for organic matter content analysis (~10 g). Samples were dried at 60 °C for 2–3 weeks and then weighed for dry mass, which was divided by the sample core volume to calculate bulk density. Approximately 1 g dry sediment was subsampled, weighed, and combusted in

a muffle furnace for 5 h at 500 °C. Sample mass was measured after combustion, and loss on ignition was calculated to estimate percent organic matter content.

The top 5 cm of sediment cores was sampled for grain size analysis. Samples were shipped to Eurofins Calscience Inc. and analyzed using laser light scattering to determine percentage composition by mass of clay (<0.00391 mm), silt (0.00391–0.0625 mm), very fine sand (0.0625–0.125 mm), fine sand (0.125–0.25 mm), medium sand (0.25–0.5 mm), and larger categories of grain sizes that were not detected.

Statistics

Two-way ANOVA using site type (channel versus shoal) and season as factors and one-way ANOVA using site as the factor were performed in R (R Core Team 2014) to detect significant differences using $\alpha = 0.05$. ANOVA assumptions were checked, and data were transformed to meet the assumption of normality when appropriate. Pearson’s

correlation analysis was performed in R with the corrplot package (Wei & Simko 2021). Alternative analysis using structural equation modeling (SEM) is shown in Figure S2 in Supplemental Materials.

Results

Site Characteristics

The grain size composition of the South SF Bay sediments (Fig. 3A) averaged 60.2% silt (31–68% range), 28.1% clay (11–36% range), and 5.7% very fine sand (0–13% range) across sites. S11 was distinct in composition as the only site with detectable fine sand making up 49% of total mass and the lowest percent mass of clay and silt. Shells were removed before grain size analysis, but sites S7, S8, and S9 tended to have shell debris mixed in the surface layer. Sediment organic matter (OM) ranged from 3 to 9% (Fig. 3B), with the lowest percent composition at sandy site S11; the channel sites had greater OM content than shoal sites ($p < 0.001$). OM at C2 and C4 was significantly greater than S2, S7, and S11 ($p < 0.04$); C5 was significantly greater than S7 and S11 ($p < 0.05$); and C6 was significantly greater than S11 ($p < 0.05$). No significant differences were detected between shoal sites, except for S8 and S11 ($p = 0.02$). NO_3^- was the dominant form of dissolved inorganic N (DIN) in the water

column (Fig. 3C), but 1–15 μM of NH_4^+ was also present. The northernmost shoal sites (S7 and S8) consistently had the lowest concentrations of DIN. Phosphate was also present in the water column at 1–3 μM (Fig. 3D).

Sediment Oxygen Demand

SOD in South San Francisco Bay (Fig. 4) ranged from 225 to 3243 $\mu\text{mol O}_2 \text{ m}^{-2} \text{ h}^{-1}$. Overall, there were no significant differences between channel and shoal SOD rates, but SOD rates at S7 were statistically greater than S2 and C6, and S8 was greater than C6. February SOD rates were significantly lower than May ($p < 0.001$), August ($p = 0.006$), and November ($p = 0.02$) SOD rates. The shoal sites had statistically higher SOD rates than the channel sites only during the May sampling ($p < 0.001$).

Denitrification Rates

Denitrification rates (D14, Fig. 5) averaged $37.4 \pm 5.5 \mu\text{mol N m}^{-2} \text{ h}^{-1}$. No significant differences were detected across seasons, but shoal site D14 fluxes were significantly higher than channel site fluxes ($p = 0.0007$). Site S2 had significantly greater D14 fluxes than all the other sites ($p < 0.0001$), and site S7 had significantly greater D14 fluxes than all sites except for S2 ($p < 0.02$). Coupled nitrification–denitrification (Dn) comprised the majority

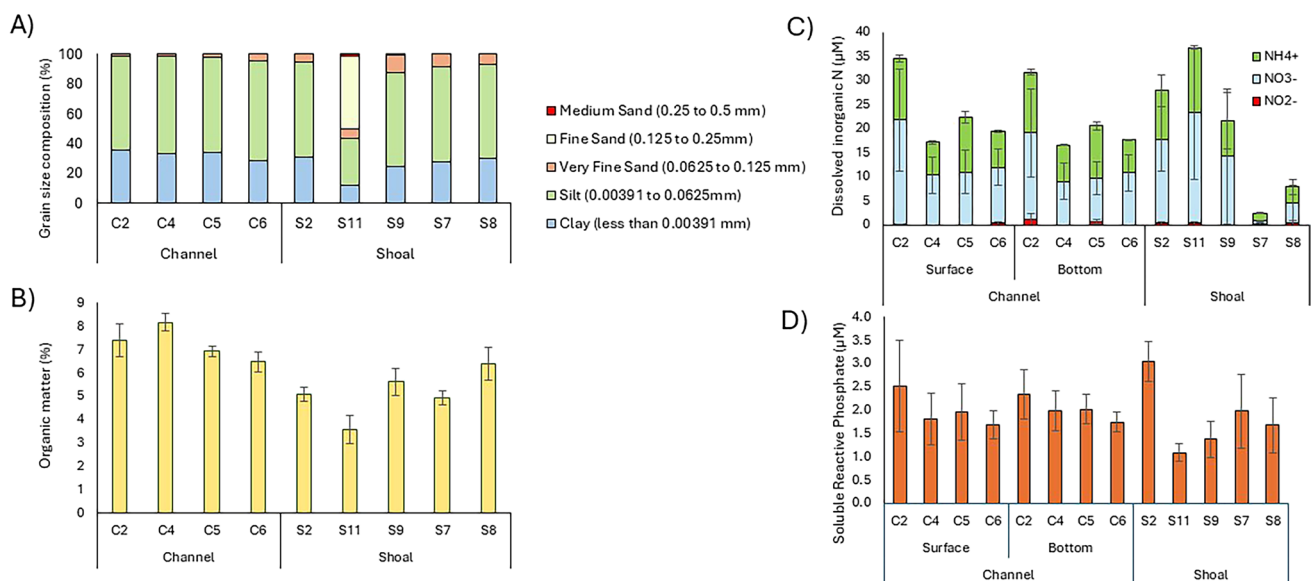


Fig. 3 **A** Grain size composition by percent mass in the top 5 cm of sediment averaged from cores from two different seasons (except for C4, which had only one core); **B** percent organic matter (0–2 cm depth) by mass as determined by loss on ignition, averaged across seasons ($n=8$); **C** average water column concentrations of dissolved inorganic N (DIN) as nitrite (NO_2^-), nitrate (NO_3^-), ammonium (NH_4^+) at each site across summer and fall seasons ($n=4$); and **D**

dissolved soluble reactive phosphate (PO_4^{3-}) at each site ($n=4$). Nutrients were measured using common colorimetric methods (Liao 2001; Gine et al. 1980; Murphy & Riley 1962). Note that NO_3^- water column concentrations differ from Fig. 2 average concentrations because these values represent two samplings of instantaneous NO_3^- that vary widely both spatially and temporally in the Bay. Sites within site type are arranged south to north. Error bars, SE

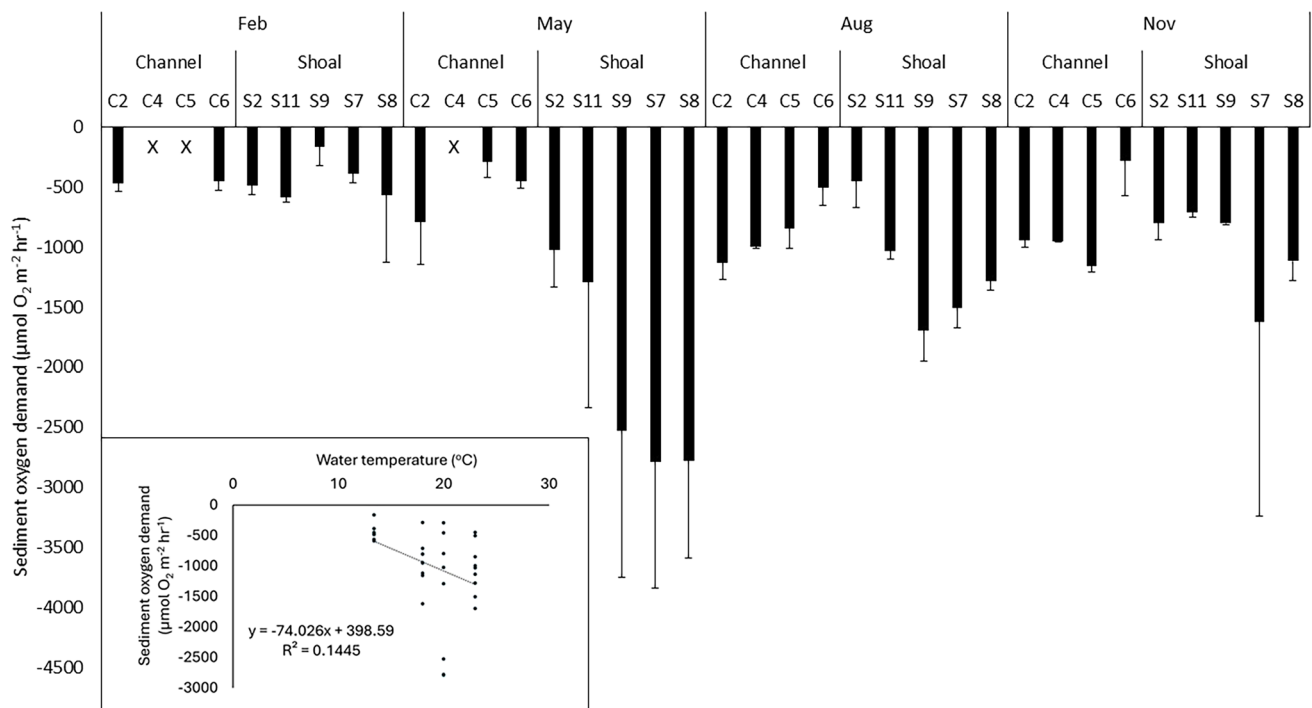


Fig. 4 Sediment oxygen demand as measured in sediment cores with a FireSting O_2 sensor. Error bars, SE. x, no data. Insert plot shows linear correlation between water incubation temperature and SOD

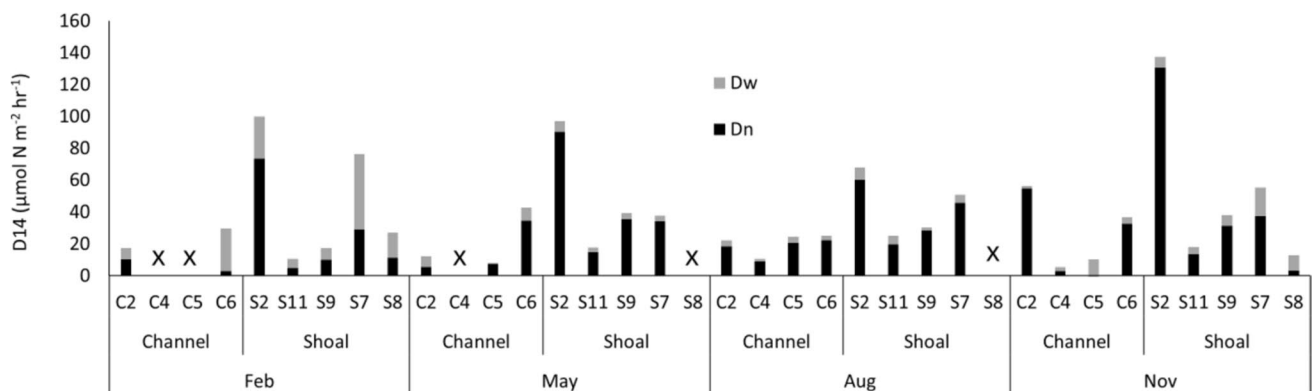


Fig. 5 Natural denitrification (D14) rates broken down into coupled nitrification–denitrification (Dn) and water column free nitrate-derived denitrification (Dw) measured in sediment cores. x, no data

(average ~70%) of D14 fluxes across most sites during all seasons. The relative contribution of Dw tended to be highest in February, coinciding with the highest concentrations of water column nitrate.

DNRA Rates

DNRA14 rates (Fig. 6) averaged $3.1 \pm 0.8 \mu\text{mol m}^{-2} \text{h}^{-1}$. Channel and shoal sites were not significantly different, but February and August DNRA14 rates were significantly greater than November DNRA14 rates ($p < 0.044$). Across

these SF Bay sites and seasons, DNRA was a minor component of NO_3^- reduction, representing on average 10% of the NO_3^- reduction across sites and seasons (Fig. 6). However, DNRA approached 50% of the NO_3^- reduction rates at S11 during February and August 2021 and at C2 during February.

Drivers of Rate Variation

Pearson's correlation analysis (Fig. 7) revealed significant positive correlations between SOD and temperature, but

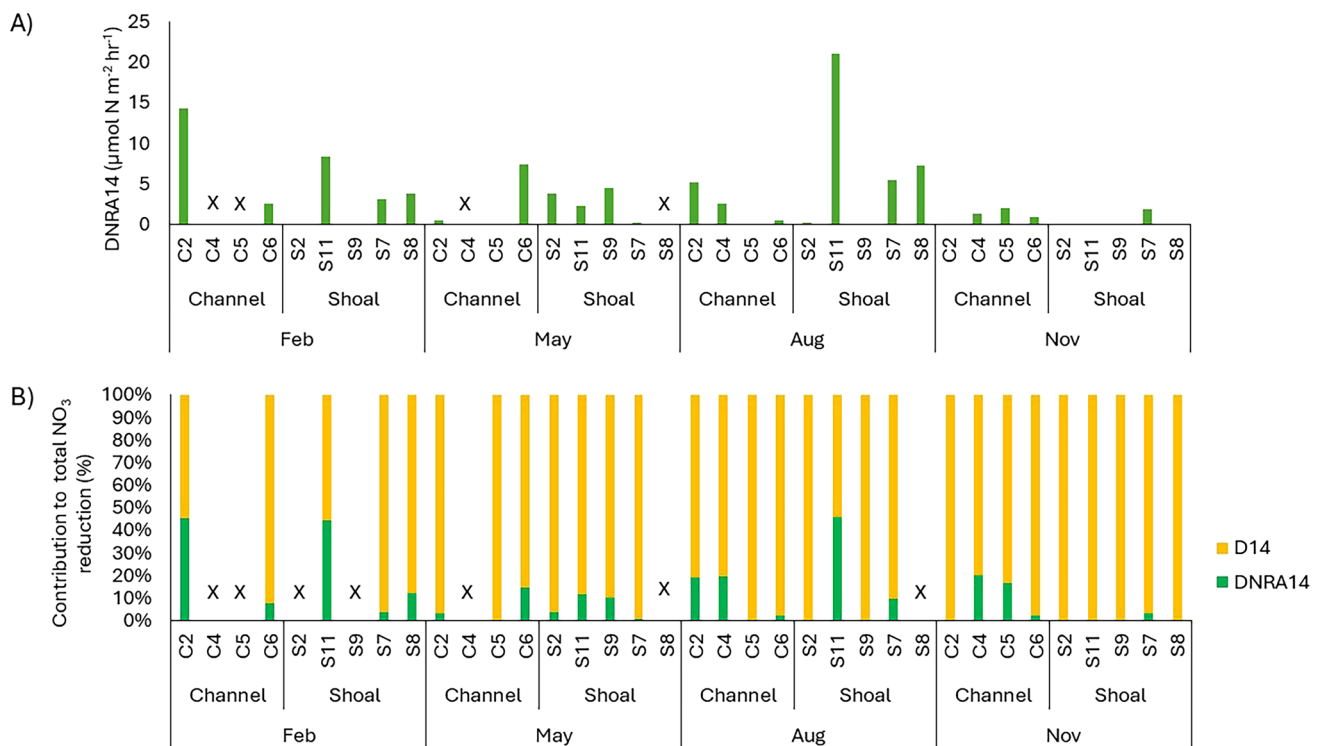


Fig. 6 **A** DNRA14 rates and **(B)** percentage of nitrate reduction as DNRA (N retention) vs denitrification (N removal). X, no data

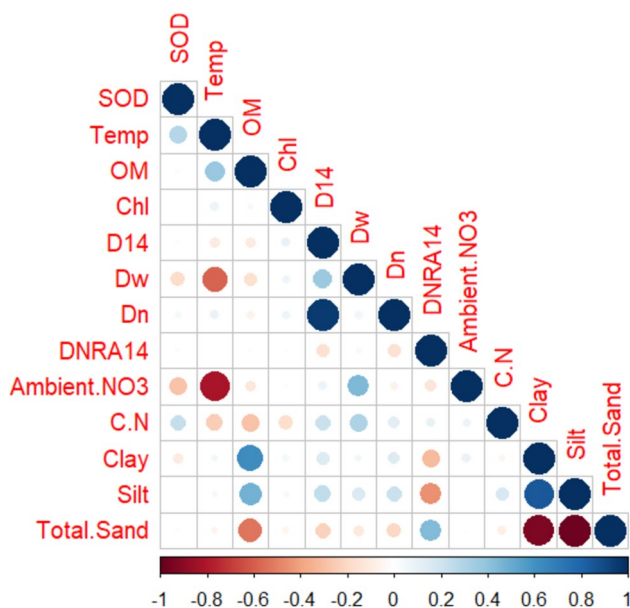


Fig. 7 Pearson's correlation matrix showing correlations between all variables measured in South SF Bay

D14 rates were not significantly correlated with any of the measured geochemical characteristics. DNRA14 was tightly positively correlated with sand content but negatively correlated with clay and silt content.

Discussion

Overall Denitrification Rate Comparison

Despite much greater N loading rates to SF Bay and higher NO_3^- concentrations in overlying water relative to other estuaries where benthic N cycling has been quantified, rates of denitrification were similar to many other northern United States estuarine systems (e.g., Massachusetts Bay, Tucker et al. 2014; Narragansett Bay, Fulweiler & Heiss 2015; Waquoit Bay, Foster & Fulweiler 2014), but lower than rates in the eutrophic mainstem of Chesapeake Bay (Francis et al. 2013) and Potomac River Estuary (Cornwell et al. 2016). A synthesis study (Fennel et al. 2009) determined that the mean denitrification rate across estuarine studies ($n = 683$) was $90 \mu\text{mol m}^{-2} \text{h}^{-1}$, a rate that is slightly higher than mean South SF Bay observations (average 60.6 ± 8.1 ; 5–137 $\mu\text{mol N m}^{-2} \text{h}^{-1}$ range). Thus, the excess NO_3^- loaded to South SF Bay did not result in greater-than-average estuarine N removal rates as we hypothesized.

The rates presented here are similar to previous observations in northern sub-embayments of the SF Bay system and other N-enriched estuarine systems. Rates of N_2 -N efflux from northern San Francisco Bay in 2011/2012 using N_2 :Ar gas ratios (Cornwell et al. 2014) ranged from 3 to 54 $\mu\text{mol N m}^{-2} \text{h}^{-1}$ in March and 9 to 97 $\mu\text{mol N m}^{-2} \text{h}^{-1}$ in September, with mean rates in March and September of 18 ± 8 and

$33 \pm 13 \mu\text{mol N m}^{-2} \text{h}^{-1}$, respectively. Although in a different US region and lower latitude, high N loading into Weeks Bay (AL, USA; Domangue & Mortazavi 2018) similarly did not result in high denitrification rates, with DNRA dominating over denitrification and likely promoted by high sulfide concentrations. Similarly, Wadden Sea (Germany, EU) had mean rates of $49 \mu\text{mol N}_2 \text{m}^{-2} \text{h}^{-1}$ (sand) and $159 \mu\text{mol N}_2 \text{m}^{-2} \text{h}^{-1}$ (finer sediment) (Deek et al. 2012). In the Tomales Bay estuary, just north of San Francisco Bay, Joye et al. (1996) estimated higher denitrification rates than those in the SF Bay in March and September 1992 of 75 ± 38 and $388 \pm 192 \mu\text{mol N m}^{-2} \text{h}^{-1}$, respectively, using a modified acetylene-based technique. This Tomales Bay study showcases a nearby estuary with more traditional estuarine characteristics and higher denitrification rates than those found in South SF Bay despite lower N loading rates.

We found that denitrification rates (D14) were greater in shoal sites than channel sites overall, and the two sites with consistently greatest D14 rates were shoal sites S2 and S7, suggesting that some shoal areas across the Bay may contain hotspots for denitrification. Lower South Bay, the sub-embayment where S2 is located, likely has greater N removal rates than South Bay due to a large proportion of shoal area. Despite site S11 having the most unique sediment characteristics of the nine sites (higher sand content, lower clay content and % organic matter), this difference in sediment characteristics did not result in significantly different biogeochemical flux rates. We did not observe strong upstream to downstream spatial patterns except for the upstream end-member shoal site (S2) having the highest D14 rates across all seasons. While we expected to measure greater N removal in deep channels (resulting from stratification, oxygen depletion and possibly more organic deposition in deeper waters), the channel within South SF Bay is typically well-mixed by strong tidal energy and oxygen levels remain high near the benthos. Despite high availability of DIN in South SF Bay, only modest rates of sediment denitrification were observed. This suggests that rates of denitrification are limited by something other than water column NO_3^- availability, such as carbon availability (Teixeria et al. 2010) or diffusion rates of oxygen and substrates into sediments (Gongol & Savage 2016). Denitrification in South SF Bay may be limited by (1) low rates of algal production and organic matter inputs resulting in limited OC substrates, (2) low rates of ammonium mineralization and nitrification, and/or (3) low influx of overlying water nitrate. Although it may appear counterintuitive that a system with high water column NO_3^- concentrations would rely primarily on nitrification for its supply of nitrate (as we observed here), this may be evidence for limitation of NO_3^- diffusing into the anoxic sediment layer, and/or the NO_3^- supply through nitrification exceeding the rate of supply by diffusion. Other estuarine benthic sediments show similar dominance of coupled

nitrification–denitrification, although the water column often becomes the main source of NO_3^- for denitrification at concentrations over $60 \mu\text{M NO}_3^-$ (Seitzinger et al. 2006).

Denitrification rates in heavily studied estuaries such as the Chesapeake Bay are thought to be controlled by NO_3^- or oxygen supply (Kana et al. 1998; Cornwell et al. 1999; Rysgaard et al. 1994). Labile organic matter inputs and a supply of nitrate are essential to create a sufficiently reducing environment to support benthic denitrification. In some cases, an oversupply of organic matter may shift overlying water-derived nitrate reduction to DNRA (Burgin and Hamilton 2007; Dong et al. 2011; Marchant et al. 2016), particularly with high rates of sulfate reduction (Giblin et al. 2013). In addition, high inputs of labile organic matter, especially in conjunction with water column oxygen depletion, can limit oxygen penetration into the sediment and limit both nitrification and subsequent denitrification (Owens & Cornwell 2020). With high concentrations of overlying water nitrate and dissolved oxygen, typical of SF Bay, organic matter inputs are likely the main limits to microbial denitrification rates.

DNRA Rates

DNRA, a key anaerobic nitrate reduction pathway often ignored in estuarine nutrient balances, may impact denitrification rates by competing for NO_3^- . Although NO_3^- concentrations are high in the water column, most of the NO_3^- used for denitrification and DNRA observed in the South San Francisco Bay sediments was supplied by nitrification. Whereas DNRA has been shown to out-compete denitrification for NO_3^- at higher temperatures in estuarine sediments (Gruca-Rokosz et al., 2009) and wetlands (Rahman et al. 2019), our observation of $0\text{--}20 \mu\text{mol N m}^{-2} \text{h}^{-1}$ DNRA representing $\sim 10\%$ of NO_3^- reduction in South SF Bay is comparable with a previous study in another temperate estuary, Niantic River Estuary, CT, which determined that about 90% of nitrate was reduced by denitrification, while the remaining 10% was reduced by DNRA or anammox pathways (Plummer et al. 2015). Similar to South SF Bay, DNRA in the Yarra River estuary in Australia averaged $28 \mu\text{mol m}^{-2} \text{h}^{-1}$, representing 18% of total NO_3^- reduction (Roberts et al. 2014); in contrast, a shallow lagoonal estuary in Texas had potential DNRA rates up to $99 \mu\text{mol m}^{-2} \text{h}^{-1}$ and DNRA represented the majority of total NO_3^- reduction (Gardner et al. 2006). The lack of strong upstream–downstream patterns of process rates often observed in other estuaries (Smith et al. 2015; Wei et al. 2020), aside from the greater denitrification rates in the LSB shoal, may be a result of South SF Bay's lack of a classical estuarine gradient in salinity and sediment characteristics.

Drivers of N Cycling Process Rates

For South SF Bay, Pearson's correlation analysis revealed a weak positive correlation between potential denitrification and SOD. SOD has been observed to be positively correlated to denitrification rates across several studies and coastal ecosystems (Piehler & Smyth 2011; Seitzinger & Giblin 1996). DNRA15 rates were driven by % organic matter but negatively impacted by % clay content. Pearson's correlation analysis revealed that grain sizes are tightly correlated with DNRA rates but not as tightly correlated with denitrification rates. The negative correlation between DNRA and clay content appears counterintuitive because DNRA is thought to be favored in locations with greater organic matter which is often correlated with clay content. However, grain size composition also controls diffusion rates into the sediment, potentially causing differences in electron acceptor (DO , NO_3^- , SO_4^{2-} , etc.) penetration depths and controlling NH_4^+ mineralization and nitrification rates. Higher clay content may adsorb more NH_4^+ and reduce DO diffusion rates, limiting sediment nitrification rates. Thus, lower clay content may allow increased nitrification, providing more NO_3^- substrate for DNRA.

D14 rates were not strongly correlated with any sediment characteristics, but highly correlated with Dn (coupled nitrification–denitrification rates), suggesting the high importance of coupling to nitrification. Correlations between D14 and grain sizes were weak but positive for clay and silt but negative for sand. Positive correlations between clay content and denitrification have previously been observed in Yangtze Estuary (Liu et al. 2019). Sediment OM content has been shown to drive rates of denitrification in a microcosm experiment (Caffrey et al. 1993), conflicting with what we have observed in South SF Bay. Whereas one would expect organic matter to provide C substrates and NH_4^+ through mineralization to support denitrification, the lability of this organic matter is unknown and much of the OM may be

recalcitrant as a result of low productivity in the Bay. The negative effect of organic matter content on denitrification may be related to competition for NO_3^- substrates with DNRA, because DNRA tends to be favored over denitrification in environments with high OM: NO_3^- ratios (Burgin & Hamilton 2007). Because DNRA competes for NO_3^- substrates with denitrification, one would expect higher DNRA rates to lead to lower denitrification rates. However, in a scenario where denitrification is tightly coupled to nitrification and nitrification rates may be limited by the supply of NH_4^+ , coupled nitrification–denitrification may rely on NH_4^+ recycled by DNRA.

Nitrogen Loading Versus Removal Rates

Seasonal N loading (Table 1) normalized to embayment surface area was highest in the winter. Thus, during winter a greater proportion of N (> 99%) was available in the Lower South Bay for alternate fates such as export to the South Bay or burial. Lower South Bay had greater N loading rates and removal rates per square meter than South Bay, but South Bay removal rate as a percentage relative to the N loading rate was over twice as large as the removal rate percentage in the Lower South Bay (~6 to 62% vs ~3 to 8%, respectively). Our data suggest that a wide range, with a maximum of 62%, of N loaded to the South San Francisco Bay may realistically be removed via denitrification at current N loading rates. This comparison only considers anthropogenic loading rates from POTW point sources but not deposition, runoff, or other alternate sources of N loading. In the northern San Francisco Bay Delta region, Cornwell et al. (2014) estimated that ~10% of nitrogen loading from the Sacramento River and San Joaquin Rivers was denitrified. The average removal rate of 14% of the N loaded to the South Bay is also strikingly similar to the assessment of 14% of N removed from the Elbe River in the Wadden Sea (Deek et al. 2012),

Table 1 N loading and removal rate seasonal comparison. N loading rates are averaged over the years 2012–2020 $\pm$ SE. N removal means and ranges were derived from IPT rates. N loading standard error represents interannual variability, and N removal ranges were derived

from the minimum and maximum rates observed at any site within the embayment during the specified season. N loading data obtained from Novick and Senn (2014) and Bay Area Clean Water Agency (BACWA) Group Nutrient Annual Reports

Embayment	Month	N loading ($\text{g m}^{-2} \text{ day}^{-1}$)	N removal (D14, $\text{g m}^{-2} \text{ day}^{-1}$), mean and (range)	% loaded N removed, mean and (range)
Lower South Bay (30 km^2)	February	0.44 ± 0.018	0.020 (0.0058–0.034)	4.51 (1.33–7.70)
	May	0.37 ± 0.0068	0.018 (0.0041–0.033)	5.05 (1.14–8.97)
	August	0.32 ± 0.0070	0.015 (0.0075–0.023)	4.84 (2.38–7.30)
	November	0.40 ± 0.0067	0.033 (0.019–0.046)	8.21 (4.77–11.66)
South Bay (250 km^2)	February	0.033 ± 0.0013	0.011 (0.0068–0.028)	15.05 (10.59–78.62)
	May	0.030 ± 0.00059	0.008 (0.0027–0.014)	14.41 (8.74–47.07)
	August	0.030 ± 0.00067	0.003 (0.0003–0.017)	14.43 (12.14–57.30)
	November	0.029 ± 0.00051	0.006 (0.0018–0.019)	13.18 (6.18–64.75)

previously mentioned for its similarity in high N loading and low denitrification rates.

Seitzinger et al. (2006) suggest that more than 50% of N loaded to coastal waters may be removed via denitrification on a global scale, and our estimate suggests that an environmentally important proportion of N loaded to South San Francisco Bay may be denitrified at the local scale. However, a significant proportion of N is also either exported to the ocean or buried in SF Bay sediments. Low primary production rates (Cloern et al. 2020), and therefore low rates of assimilation and sinking of organic N, suggest that burial would not likely be the fate of a significant proportion of DIN.

Conclusions

Research on microbial nitrogen cycling in estuaries is largely biased toward a few heavily studied systems that are highly impacted by eutrophication. Greater understanding of N cycling dynamics is needed in atypical estuaries that do not fit classical biogeochemical eutrophication paradigms. Here we have shown nitrogen cycling and removal dynamics in a large, understudied estuary that has several unusual features, such as high turbidity and low productivity, which buffer this system from the negative impacts of eutrophication. Despite the unusually high N loading rate to South San Francisco Bay, denitrification rates were well within the typical range observed across other estuaries with much lower N loading rates and water column concentrations. The greatest denitrification rates were observed at shoal sites rather than channel sites, revealing hotspots of denitrification in the broad shoal areas that represent the majority of the benthic surface area in the Bay. Channel denitrification rates may have been low as a result of well-mixed waters and high oxygen concentrations even at the deepest channel sites, as well as limitation of labile organic C resulting from low productivity. While we did not observe the expected strong upstream to downstream patterns in N process rates, likely due to a lack of strong gradients in characteristics such as salinity, nutrients, or sediment grain size, Lower South Bay did have greater denitrification rates (D14) on the shoal than the South Bay. DNRA rates comprised about 10% of total NO_3^- reduction rates. Denitrification may remove about 14% of N loaded to the South San Francisco Bay with a broad range of 2–62%, suggesting that it plays an important but not a dominant role in the fate of N loaded to the Bay. These findings represent a first step in determining N cycling rate magnitudes, spatial and seasonal variation, and drivers within South SF Bay, leading to future work such as modeling denitrification rates and characterizing sediment microbial communities responsible for these N cycling characteristics. This work fills a major data gap for biogeochemical models of SF Bay

and provides greater understanding of the Bay's unique N cycling dynamics and removal of N inputs, which is critical for management and prediction of future changes to water quality. This also provides a well-timed baseline prior to the August 2022 South SF Bay harmful algal bloom event.

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Data Availability The data can be made available upon request.

Declarations

Competing Interests The authors declare no competing interests.

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